

Paediatric (Childhood) Acute Stroke Protocol & Guide

1. Definitions, Epidemiology, Outcomes

Acute stroke is defined as an acute onset neurological sign or symptom attributable to focal brain infarction or haemorrhage. Acute stroke in children can be classified into neonatal and childhood strokes. Neonatal strokes, or **perinatal strokes**, are defined as stroke occurring from 28 weeks' gestation to 28 postnatal days of life. **Childhood strokes** occur after 28 days to 18 years of age. Stroke can be ischaemic or haemorrhagic; ischaemic stroke includes arterial ischaemic stroke (AIS) and venous infarction caused by cerebral venous sinus thromboses (CVST). **For the purposes of this protocol, the focus will be on childhood stroke.** Childhood strokes are rare at 2-2.7 per 100,000 per year. AIS accounts for approximately 50% of childhood strokes. Most of the remaining cases constitute haemorrhagic stroke; cases of CVST are rare at 0.3 per 100,000 per year. The incidence of ischaemic strokes is highest in infants and children who are < 5 years old, more so in boys than girls, more so in Asian and African American ethnic groups. Overall mortality in childhood strokes is about 4-5%; 70% of those who survive have long term neurological deficits (36% mild, 23% moderate, 10% severe) and have overall reduced cognitive function (difficulties with inattention, impulsivity, executive function).

Why is this protocol/guide important?

It is important to have a well-coordinated and aligned multidisciplinary medical care plan the moment a healthcare practitioner suspects acute stroke. Time is Brain!

2. Childhood stroke: Causes and Clinical Presentation

i. Causes and Clinical Presentation

The common signs and symptoms of childhood stroke include: hemiparesis and hemifacial weakness (67-90%), speech disturbance (20-50%), vision disturbance (10-15%), ataxia (8-10%). These may occur alongside non-localizing symptoms like headache (20-50%) and altered mental status (17-38%). Seizures at presentation are more common in children <6yo than in adults with AIS (15-25%, compared with 2.4-8% in adults). Compared to AIS and haemorrhagic strokes, CVST tends to have a more subacute or chronic presentation: headache, nausea and vomiting, lethargy, diplopia (or proptosis in cavernous sinus thrombosis). This corresponds to the gradual buildup of venous pressure resulting in obstruction of arterial flow. Of the various forms of stroke, AIS and haemorrhagic stroke require urgent evaluation and treatment, while CVST management is mostly supportive.

Childhood ischaemic and haemorrhagic strokes have different predisposing factors/conditions.

Childhood AIS is mostly related to cardioembolic phenomena, arteriopathy (eg. Moya Moya), thrombophilia, inflammatory vasculitis, and genetic/metabolic conditions – summarized in **Table 1 (Pg 7-8)**. **Childhood haemorrhagic strokes** are caused by structural lesions in 75% of the cases (AVM being the most common) – summarized in **Table 2 (Pg 8)**. **Childhood CVST**, more rarely seen, is often unique to patients with comorbidities predisposing to thrombus formation – CNS, head and neck infections, nephrotic syndrome, inflammatory conditions, oncological conditions as well as medication intake (L-

asparaginase, oral contraceptives). Please refer to Tables 1 and 2 for a comprehensive but non-exhaustive summary of the causes and features of both AIS and childhood haemorrhagic stroke.

ii. Mimics of Childhood Stroke

Because these mimics are often associated with neurological complaints, there should be a low threshold for brain imaging in each case. Up to 40% of children with stroke may have an incorrect initial diagnosis of a stroke mimic at presentation. This may be related to the spectrum of signs and symptoms seen in children with acute stroke. Focal deficits are more common in children with stroke than in those with stroke mimics; non-localizing symptoms such as headache and AMS are equally common in both groups. Clinical diagnostic tools in adults to distinguish stroke mimics have limited utility in children, with a sensitivity of 60%.

- a. Migraine with aura
- b. Bell's palsy
- c. Seizure with Todd's paresis
- d. Brain tumour
- e. Demyelinating disease
- f. Cerebellitis / encephalitis
- g. Epidural abscess
- h. Traumatic brain injury
- i. Intoxication
- j. Metabolic disease
- k. Psychogenic disorders

3. Useful Evaluation Tools

i. Imaging

In general, Computed Tomography (CT) helps to look for haemorrhage. Magnetic Resonance Imaging (MRI) can show cerebral ischemia and edema in dedicated sequences. Magnetic Resonance Angiogram (MRA) helps to delineate arterial vascular anatomy. MRI and MRA can be useful if the time of onset of deficit is unclear. However, MRI takes time and should not delay treatment if the diagnosis is clear. Magnetic Resonance Venography (MRV) will be useful in CVST. In the less acute phase, digital subtraction angiography (DSA) is the gold standard to study the structure of cerebral blood vessels if there is suspicion of arteriopathy. **In our centre, CT + CTA would be the most practical and efficient first step when acute stroke is suspected.**

ii. Clinical examination

Focal neurological deficits: hemiparesis/hemifacial weakness, aphasia, visual disturbances, ataxia; **± any:**

- a) Non-localizing symptoms: headache, altered mental status
- b) Raised intracranial pressure: Cushing's reflex, Lateral gaze palsy, anisocoria, papille dema
- c) Evidence of trauma / ICH: Scalp haematoma, raccoon's eyes, epistaxis, haemotympanum

iii. PedNIHSS score

National institute of Health Stroke Scale (NIHSS) is a reliable quantitative measure of acute stroke severity in adults, predicting stroke outcome at 7 and 90 days. The modified PedNIHSS allows one to

assess children in an age-appropriate manner and has demonstrated good inter-rater reliability in ages 2-18 years. It tabulates a score that ranges from 0 to 42 (the higher, the worse the prognosis). Its components assess the level of consciousness, gaze palsy/visual field defects, motor deficits, sensory deficits, language deficits and neglect. **This scale helps to stratify the severity and prognosis of each case, hence guiding acute treatment decisions. A score that signifies at least moderate deficit would fulfil criteria for consideration of recanalization therapy.** This score also helps to quantify outcomes after recanalization therapy. **The PedNIHSS is to be used in the activation pathway and can be found in our department guidelines under “Paediatric Critical Care” or “Children’s Emergency”.**

4. Principles of Management

4.1. Acute Resuscitation

a) Airway management – Consider intubation if

- Evidence of raised ICP
- Altered GCS ≤ 8 or fluctuating GCS
- Status epilepticus
- Unable to maintain airway

b) Consider general neuroprotective measures

- Nurse head up 30 degrees whenever possible
- Normothermia (cooling blanket should be ready before patient touched down in PICU)
- Avoid hypotension or blood pressure $< 50^{\text{th}}$ centile
- Na aims 140-145mmol/L, normocarbica 35-45mmHg
 - o *If there are signs of intracranial hypertension, please refer to Paediatric Critical Care guidelines: “Management of Paediatric Traumatic Brain Injury” for CPP targets.*
- 3% NaCl 3-5ml/kg, Mannitol 20% 2ml/kg

4.2. Establish diagnosis (Stat CT brain $\pm$ CTA)

1. Note time of onset of symptoms and PedNIHSS.

2. Consider a stat CT Brain and CT Angiogram (CTA) and **inform Paediatric Neurology early.**

3. If a stat CT brain shows cerebral haemorrhage, inform the **Neurosurgeons immediately.**

4.3. Specific types of stroke

i. Acute Ischaemic Stroke (AIS)

a. Hyperacute recanalization therapy (IV thrombolysis, mechanical thrombectomy)

Evidence for IV thrombolysis is limited to observational studies. A retrospective review (Amlie-Lefond et al. 2020) done on 26 patients aged 1-17 years old showed no cases of symptomatic intracranial haemorrhage (SICH) after receiving IV TPA within 4.5 hours from symptom onset. Also, median risk estimates of SICH were 2-4%, which is lower compared to that in adults receiving IV TPA for AIS (6%).

There is growing evidence for mechanical thrombectomy in large vessel occlusion in paediatric AIS. In children aged 1-18 years, mechanical thrombectomy has been found to be safe with 87% recanalization

rates, with significant short- and long-term neurologic improvement from presentation (*Save ChildS Study, JAMA Neurology 2020*). Post interventional SICH was low at 1%. Comparing against findings from a large study on adult AIS patients receiving mechanical thrombectomy (HERMES meta-analysis), the proportion of children with SICH was lower; positive neurologic outcomes were also comparable. There are other small case series that support these findings.

According to the 2019 AHA statement on the management of paediatric strokes and various sources, **thrombolysis / thrombectomy can be carefully considered in a select group of patients and should be discussed with the Paediatric Neurology, Neurosurgery and Paediatric ICU:**

- a. Diagnosis of ischemic stroke established in 2-18 years old*
- b. Within therapeutic window (4.5hrs for thrombolysis, 6hrs for thrombectomy)*
- c. pedNIHSS score ≥ 6 points, ≤ 24 points at the time of intervention*
- d. Radiographically confirmed cerebral large artery occlusion correlating with neurological deficit*

Contraindications to hyperacute recanalization therapy

- a. Thrombolysis
 - i. Brain/other vascular malformations
 - ii. Brain tumour
 - iii. Congenital or acquired bleeding diatheses
 - iv. Thrombocytopenia
 - vi. Any thrombolysis / anticoagulation administered in the last 24-48 hrs
 - vii. Gastrointestinal / genitourinary tract bleeding in the preceding 21 days
 - viii. Haemorrhage seen on CT Brain/MRI Brain, or haemorrhage elsewhere
 - ix. Extensive infarct size on initial CT scan
- b. Endovascular Thrombectomy – EVT (relative)
 - i. Small child with accompanying technical limitations
 - ii. Suspicion of underlying vasculitis (eg. FCA) or arteriopathy (eg. Moyamoya)
 - iii. Serious sensitivity to radiological contrast/nickel, titanium metals or their alloys
 - iv. Significant acute kidney injury / chronic kidney disease (risk of contrast-induced nephropathy)

b. Prevention of secondary brain injury and recurrent strokes in AIS

- a. Normoglycemia, normothermia and normotension

Due to the lack of paediatric data, it is reasonable to follow AHA/ASA adult stroke guidelines to maintain normoglycemia, normothermia and normotension. If there is hypertension, it would be reasonable to lower BP by 15% during the first 24 hours, but cautiously, given that brain ischemia can be exacerbated in vascular stenosis (eg. Moyamoya), or in cases of raised intracranial pressure.

- b. Seizure management/prophylaxis

Clinical seizures occur in at least 20% of children with AIS. Patients will benefit from close monitoring of clinical and electrical seizures via continuous EEG. Consider seizure prophylaxis (Keppra 10mg/kg BD).

c. Anti-platelet/anti-coagulation

Anti-platelet (Aspirin) or anti-coagulation therapy (LMWH) appear to be safe in initial AIS. There is significant risk of stroke recurrence (10%) within a year of the index stroke, with the highest recurrence risk being 30% in those with progressive arteriopathies. This risk is doubled when anti-platelet or anti-coagulation therapy is not initiated. Aspirin or anticoagulation may be initiated in the initial 5-7 days, and the regime refined after the cause is known (eg. Preference for anticoagulation if there is underlying cardiac embolism or known prothrombotic disorder).

ii. Haemorrhagic stroke

Specific treatments for vascular anomalies such as Arteriovenous Malformation (AVM), arteriovenous fistula (AVF) and aneurysms require the expertise of the intervention neurologists and neurosurgeons. Options include surgical resection, radiation therapy or endovascular techniques. Multidisciplinary discussions should be undertaken to discuss possible options. A shared decision should be made with the patient and/or family.

iii. CVST

Based on adult CVST clinical trials, there is clear benefit for **anticoagulation therapy in those with chronic conditions with hypercoagulable state** (oncological conditions, nephrotic syndromes, inflammatory conditions). **However, there is no evidence for anticoagulation in septic CSVT related to head and neck infections**, in which antibiotic therapy and surgical interventions are the mainstay of therapy – recanalization rates after mainstay therapy are high, regardless of whether anticoagulation is used. Treat the other modifiable causes/risk factors, such as intravenous fluids for dehydration.

Paediatric Stroke: Assessment and Activation Pathway in ED

Focal neurological deficits: hemiparesis/hemifacial weakness, aphasia, visual disturbances, ataxia, $\pm$ any:

- Non-localizing symptoms:** headache, altered mental status
- Raised intracranial pressure:** Cushing's reflex, Lateral gaze palsy, anisocoria, papilledema
- Known conditions:** Cardiac disease, cerebral arteriopathy, autoimmune disease

Urgent Assessment and Stabilization

- Neurologic assessment with **PedNIHSS** (under "Children's Emergency" or "Paediatric Critical Care").
- Obtain venous access (minimum 22G):
POCT glucose, micro-electrolytes
FBC, DIC screen, GXM, Urea / Cr / Electrolytes
Other invx depend on clinical concerns (if there is suspected ICH, refer below right)
- Medical stabilization
Secure airway if needed
Neuroprotective measures
- Consider nasogastric tube $\pm$ Urinary catheter

Paediatric Stroke Team*

- Paediatric Neurologist on call
- Adult Acute Stroke Team on call
- Paediatric ICU con on call

Suggested communication:

"This is a Paeds AIS stroke activation. PedNIHSS score is _____. CTB and CTA ongoing / CTB and CTA neg / concerning for ischemic stroke."

For EVT:

- Interventional neuroradiologist (Dr Anil Gopinathan 97296614/Dr Yang Cunli 91595700)

- Record time of onset of symptoms
- Record PedNIHSS score
- Activate Paediatric Stroke Team***
- Stat CT brain + CTA

Discuss with Stroke team re: MRI/MRA if time of onset is unclear

**No Intracranial Haemorrhage
Suspect ischemic stroke**

Clinical criteria:

- ≥ 2 yo
- Neurologic deficit + pedNIHSS score ≥ 6 , ≤ 24

Neuroimaging criteria:

- No intracranial haemorrhage
- Signs of ischemia consistent with large vessel occlusion (anterior and posterior circulation territories)

Intracranial Haemorrhage

In CE

- Activate PICU, Paeds Neuro, NES
- IV Access: FBC, DIC screen, GXM, Consider ROTEM [ECMO]

In PICU, consider central access

1. Before transfusions:

- 1 x EDTA tube for DNA extraction (ice)
 - 1 x Li Hep tube: Plasma aa + ac (ice)
 - Spot urine organic acids (ice)
- Discuss with various subspecs re: i) Viral titres ii) Blood spot carnitine iii) Blood Cu and Ceruloplasmin iv) factor assays
- When blood draw is possible: Ca, Phos, ALP, PTH, Vit D if skull #

Symptom onset ≤ 4.5 h

If no contraindications,
Consider IV TPA in ED
 $\pm$ EVT #

Within the shaded portion of
this protocol, all decisions to
be discussed with Paediatric
Stroke Team*

Symptom onset >4 h, ≤ 6 h

If no contraindications,
consider EVT #

Neuroprotective measures
Seizure prophylaxis (Keppra 10mg/kg BD)
PICU admission

Symptom onset >6 h

Consult stroke team

IF NO BLEED + not for TPA/EVT, consider:

- ASA 5mg/kg after 24 hours
- Neuroprotective measures
- Seizure prophylaxis Keppra 10mg/kg BD
- PICU admission

Refer to "Acute Paediatric Stroke Activation Pathway in ED" for detailed procedures and contraindications (EVT, TPA)

Appendix

Table 1: Aetiologies of childhood AIS (More common causes in YELLOW).

Category	Common causes		Mechanism	Unique clinical features	Evaluation
Cardioembolic (35%)	1. Congenital heart disease (19-fold ↑ risk) 2. Acquired heart disease 3. ECMO (Prevalence: 7-11%) 4. Arrhythmia 5. Procedures (eg. Fontan, Cardiac cath) Less common: Hereditary hemorrhagic telangiectasia (HHT)		Thromboembolic phenomena , contributed by i. Anomalies of head and neck vasculature ii. Venous stasis iii. Paradoxical venous embolism in CHD iv. Arrhythmia v. Pulmonary level shunting (HHT)	HHT – Frequent nosebleeds, abnormal blood vessels over the face, hands *27% risk of stroke recurrence in those with CHD, despite anticoagulation	1. TTE with bubble study 2. ECG and telemetry
Arteriopathy (50-60%)	Intracranial (45% of all strokes)	Moyamoya (10% of all strokes and TIA)	Progressive stenosis of distal MCA / proximal ICA with collaterals (often bilat)	Usually present as anterior circulation strokes, with suggestion of previous episodes of ischaemia *Moyamoya: 1-year recurrence risk of 35% *FCA: 1-year recurrence risk of 19-25%	CT Head and neck with CTA OR MRA Brain and neck Moyamoya/CCAD: DSA as gold standard Moyamoya: genetic testing (RNF213 mutation) FCA-i: Blood + CSF eval for VZV, HSV FCA-i / CCAD: Consider underlying CTD
		Focal Cerebral Arteriopathy (FCA) inflammatory type (FCA-i) (FCA-i <i>ctd</i>)	Usually infectious /para-infectious process → localized vessel inflammation of distal internal carotid artery and its branches → secondary thrombus formation		
		FCA dissection type (FCA-d)	Intracranial arterial dissection of the anterior circulation, typically with trauma		
		Fibromuscular dysplasia (RARE, <1%)	Idiopathic, non-inflammatory fibromuscular dysplasia	Refractory hypertension. Typical 'string of beads' angiography involving renal and cerebral vessels.	
	Extracranial (7.5% of all strokes)	Cranio-cervical arterial dissection (CCAD)	Tearing of the layers of the internal carotid or vertebral arteries → ↑ platelet deposition + activation of clotting cascade	Posterior circulation stroke RFs: Male, head and neck trauma / repetitive head turning, connective tissue disease (CTD)	Fibromuscular dysplasia: renal artery/temporal artery biopsy
Thrombophilia	Congenital	- Protein C, S deficiency - Antithrombin mutation	i) Dysregulation in thrombus formation ii) Impaired thrombus degradation iii) Pro-inflammatory state	Recurrent stroke Fam hx of thrombophilia in 1 st deg relative Symptoms and fam hx of autoimmune disorders	*Consult haematology. Consider: FBC, Protein C, Protein S, lupus anticoagulants, anticardiolipin antibodies, anti-β ₂ antibodies
	Acquired	- Sepsis - Anti-phospholipid syndrome			

*Extent remains controversial due to variability of testing – paediatric levels of protein C, S and antithrombin approach adult levels at various

times. Consult a haematologist.					± prothrombin G20210A mutation, FVL mutation, antithrombin mutation
Inflammatory	<u>Intracranial</u>	Childhood primary angiitis of the central nervous system (cPACNs)	Vasculitis	Personality change, headaches with stepwise deterioration	1. CSF testing for VZV, HSV infection 2. Consider Vessel Wall Imaging (VWI)
	<u>Systemic</u>	Para / post <u>Infectious</u> Eg. Varicella, Influenza, CoVID (overlapping with thrombotic complications) <u>Rheumatological</u> 1. Lupus 2. Takayasu arteritis 3. PAN, Wegener disease 4. P- and C-ANCA		AI disease: Renal disease, proteinuria, hand nodules, frequent fevers, livedo reticularis.	*Rheumatology consult 1. ESR, CRP, ANA, SLE panel 2. PAN: Tissue biopsy 3. Specific genetic tests
Genetic/ metabolic	PHACE syndrome, Fabry disease, MELAS, POLG1				Genetic consult

**If there is sufficient evidence for cardioembolic phenomena or arteriopathy, there may not be a need to evaluate for inflammatory causes*

Table 2: Aetiologies non-traumatic childhood haemorrhagic strokes - More common causes in YELLOW).

Category	Common causes		Mechanism	Unique clinical features	Evaluation
Structural causes (75%)	1. AVM (most common) - Haemorrhagic risk 6%, mortality 25%		AVM – abnormal connection between cerebral arteries and veins, bypassing capillary network	AVM: spontaneous intra-cerebral haemorrhage (ICH)	AVM, AVF, Aneurysms CT + CT-A DSA gold standard. CMs/Tumours: MRI/ CT RASA-I mutation, HHT predispose individuals to AVM and AVF
	2. AVF		AVF – arteriovenous shunting (external / internal carotid artery and VA → dural sinus)	Aneurysm: SAH (60%), IVH (10-15%), ICH (10-15%), SDH (1-3%)	
	3. Aneurysms		CM - Collection of enlarged and irregular capillaries	Clinical features of vascular birth marks / telangiectasias may clue in on genetic syndromes (Sturge Weber, HHT)	
Haematologic causes (10-15%)	Congenital	Hemophilia A/B / VWB disease (90%) Factor VII def, VKDB (3-5%)	Defects in platelet / coagulation pathway		Screen for bleeding disorders FBC, PT PTT INR
	Acquired	Idiopathic thrombocytopenic purpura (ITP)			
Idiopathic (10%)					After ruling out above causes

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